



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SECI 1143
PROBABILITY & STATISTICAL DATA ANALYSIS

GROUP ASSIGNMENT 1

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SEMESTER II (2023/2024)

ASSIGNMENT 1

Question 1

a) Population is a collection of all items of interest to a study.
sample is a subset of the population

b) i) discrete / continuous data

discrete :

- socioeconomic status (low, middle, high)
- number of siblings
- number of hours spent studying per week
- number of extracurricular activities
- marital status (married, single, divorced)

continuous :

- parental income
- GPA
- Test scores
- Distance between university and home
- Age

ii) categorical:

- Race
- gender
- Residential accommodation
- type of school attended
- marital status

numerical:

- Age
- GPA
- parental income
- number of siblings

iii) level of measurement

nominal : Race, gender, type of school attended, marital status

ordinal : socioeconomic status

interval : none

ratio : age, parental income, number of siblings,
number of extracurricular activities, hours
of studying per week, GPA

c) Primary data source

- questionnaires distributed to the faculty students to gather informations on their socioeconomic backgrounds

d) secondary data

Academic transcript, financial aid records, enrollment data

e) descriptive statistics

i) mean of the CQPN : 3.5

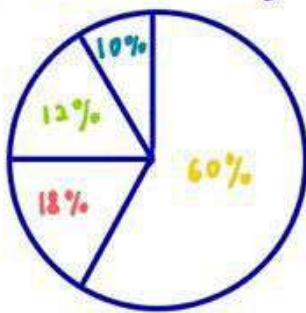
median : 3.6

mode : 3.75

Range of parental income : RM 100 000 (highest) - RM 25 000 (lowest) = RM 75 000

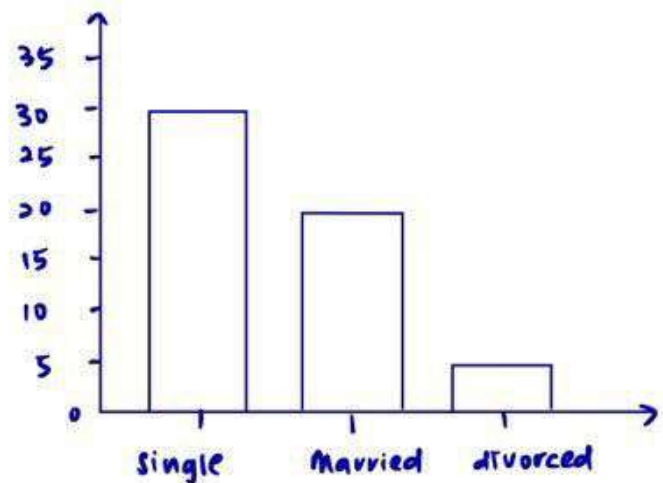
Standard deviation of CQPN : 0.7

ii) Race / Ethnicity



■ Malay
■ Chinese
■ Indian
■ others

iii) Marital Status



2.

The number of students coming in for treatment each half hour	Frequency	Relative frequency
0	3	$\frac{3}{30} = 0.10$
1	2	$\frac{2}{30} = 0.07$
2	7	$\frac{7}{30} = 0.23$
3	7	$\frac{7}{30} = 0.23$
4	5	$\frac{5}{30} = 0.17$
5	4	$\frac{4}{30} = 0.13$
6	2	$\frac{2}{30} = 0.07$
Total	30	1.00

Question 3 [25 Marks]

Construct :

a) a histogram

weight	Tabulation	Frequency	Weight	Tabulation	Frequency
450		10	462		2
451		2	463		3
452		2	464		4
453		7	465		3
454		3	466		4
455		6	467		5
456		4	468		0
457		6	469		4
458		4	470		5
459		3	471		8
460		3	472		9
461		2	473		1

②

$$\text{range} = X_h - X_l$$

$$R = 473 - 450$$

$$R = 23$$

③ cell interval

$$i = \frac{R}{1 + 3.322 \log n} = \frac{23}{1 + 3.322 (2)}$$

$$= \frac{23}{1 + 3.322 \log 100} = 3.0089$$

④

$$\text{number of cell, } h = \frac{R}{i}$$

$$= \frac{23}{3}$$

$$= 8$$

⑤ cell midpoint

$$MP_x = X_x + \frac{i}{2}$$

$$= 450 + \frac{3}{2}$$

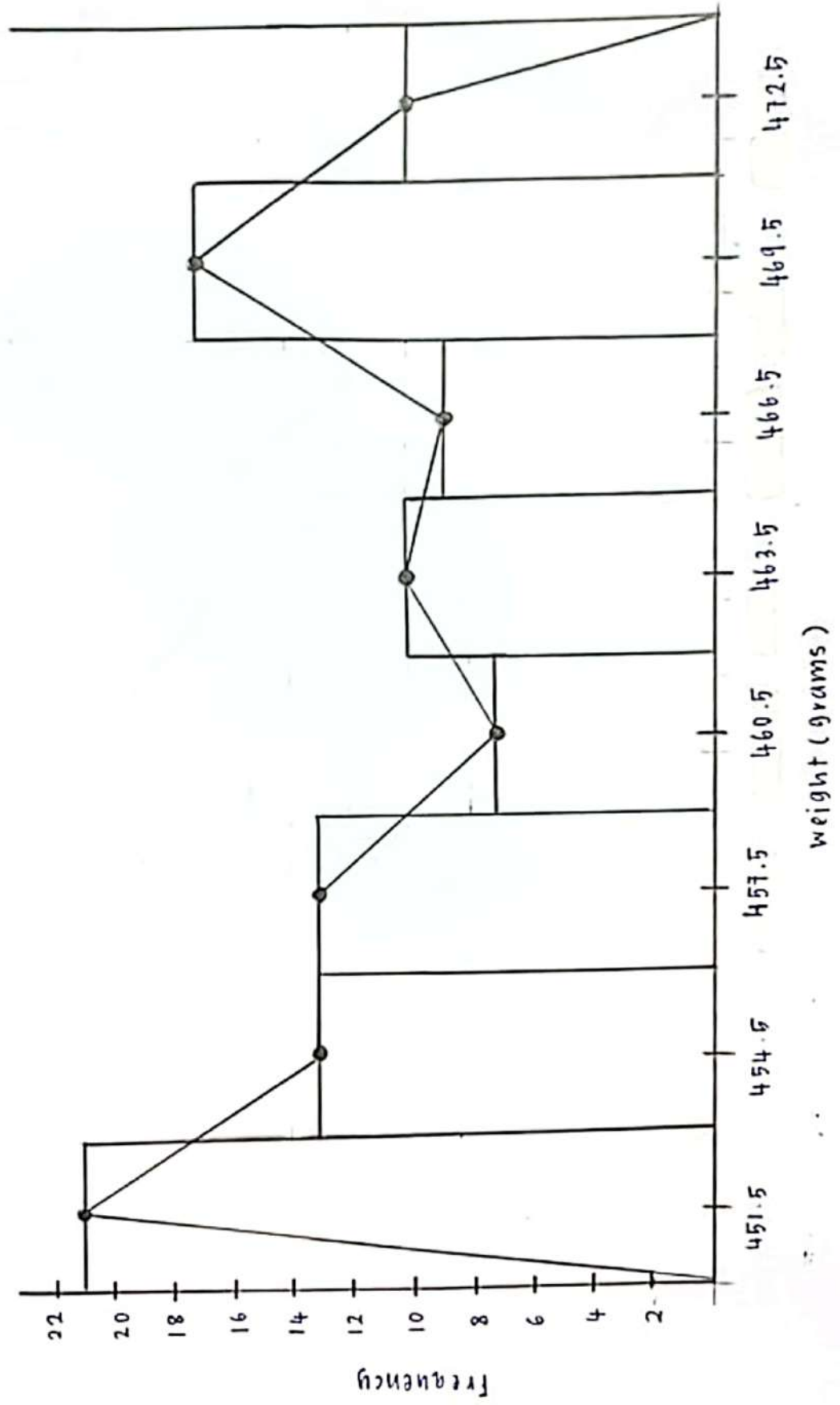
$$= 451.5$$

cell midpoint
451.5
454.5
457.5
460.5
463.5
466.5
469.5
472.5

⑥

cell boundaries	cell midpoint	Frequency
450 - 453	451.5	21
453 - 456	454.5	13
456 - 459	457.5	13
459 - 462	460.5	7
462 - 465	463.5	10
465 - 468	466.5	9
468 - 471	469.5	17
471 - 474	472.5	10

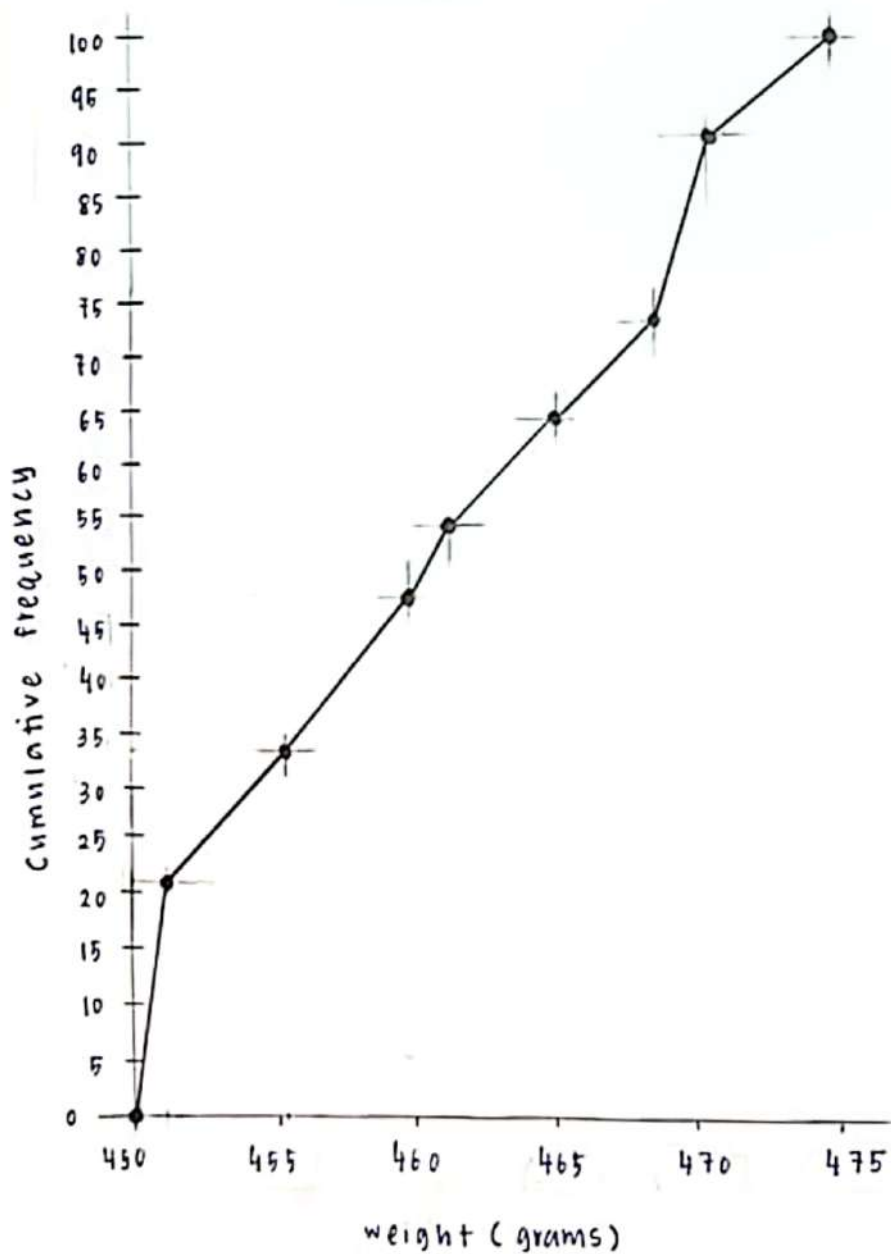
(a) Histogram and frequency polygon (b)



ogive

cell boundaries	frequency	cumulative frequency
450 - 453	21	21
453 - 456	13	34
456 - 459	13	47
459 - 462	7	54
462 - 465	10	64
465 - 468	9	73
468 - 471	17	90
471 - 474	10	100

ogive



Variables	minimum	Q1	Median	Q3	Maximum
life times	702.0	922.3	1009.0	1156.3	1340.0

The minimum value = 702

First quartile = 922.3

Median = 1009.0

Third quartile = 1156.3

Maximum = 1340

